# SAGAR KONKANAGAON

Senior Data Engineer
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## PROFESSIONAL SUMMARY

Senior Data Engineer with 5+ years building batch data pipelines on Spark and Hadoop, with production lakehouse experience on Azure Databricks and Delta Lake. Core strengths: PySpark and Spark SQL development, large-scale pipeline migration and Spark performance tuning. At eBay, on a petabyte-scale Hadoop platform with tables of ~2 trillion records, migrated 300+ production pipelines (including SLA/SLE-critical jobs) from UC4 to the in-house DLS orchestration platform and delivered multiple historical backfills. Earlier: cut Spark job runtimes by 35% on an automotive lakehouse, built schema-evolution handling for telemetry from 10,000+ IoT devices, and implemented data quality and reconciliation controls over 20M+ financial records.

## CORE SKILLS

**Core (daily production use):** PySpark, Spark SQL, Apache Spark, Hive, Hadoop / HDFS, Python, SQL
**Lakehouse & Cloud:** Azure Databricks, Delta Lake, Medallion Architecture (Bronze-Silver-Gold), Azure Data Factory (ADF), ADLS Gen2, Parquet, ORC
**Orchestration:** UC4 / Automic, DLS (Data Lake Studio), Airflow DAGs on GCP Cloud Composer
**Data Modeling & Quality:** Dimensional Modeling, SCD Type 1 & 2, Incremental Loading, Schema Evolution, Data Contracts, Data Validation, Reconciliation, Lineage
**CI/CD & Operations:** Git, Jenkins, Shell Scripting, Linux, PagerDuty
**Also worked with:** GCP BigQuery, PostgreSQL, Tableau, RAG with LLM APIs (portfolio project)

## PROFESSIONAL EXPERIENCE

### Senior Data Engineer — Jan 2026 - Present
**eBay - Bengaluru, India**
*Full-time contract engagement at eBay (payroll: Flexton)*

- **Migrated 300+ production pipelines, including SLA- and SLE-critical jobs,** from legacy UC4 (Automic) to eBay's in-house DLS (Data Lake Studio) orchestration - analysed job-plan dependencies, split large workflows into independent units and rebuilt them as DLS pipelines.
- **Develop PySpark and Spark SQL pipelines on eBay's petabyte-scale Hadoop platform**, working with tables of **~2 trillion records**, for ingestion, transformation and delivery to downstream analytical systems within data-freshness SLAs.
- **Delivered multiple historical backfills**, including a 2-year reprocess across high-volume pipelines, validating the completeness and consistency of every reprocessed dataset before release to downstream analytics.
- **Owned build and release for the ADLC migration** - built deployment artifacts, packaged and versioned JARs, and kept prior stable JAR versions as the rollback path to protect pipeline stability during cutover.
- **Validated data contracts** with the PD team before onboarding pipelines - schema, field-level definitions and source-to-target mappings agreed ahead of development.
- **Built an AI-assisted documentation repository** covering end-to-end pipeline flows, used for knowledge sharing and faster onboarding of new engineers.
- **Own production reliability for these pipelines** - root-cause Spark and Hive job failures and ship code fixes (on-call via PagerDuty).

***Stack:*** *Hadoop, HDFS, Hive, Apache Spark, PySpark, Spark SQL, Python, UC4 (Automic), DLS, ADLC, Data Contracts, PagerDuty, Git, Linux*

### Senior Data Engineer — Aug 2025 - Jan 2026
**RedBlackTree - Bengaluru, India**
*Client: Trane Technologies (HVAC / Refrigeration IoT)*

**Industrial IoT Telemetry Data Platform**

- **Designed and built Hadoop-Spark pipelines** processing HVAC telemetry from **10,000+ connected IoT devices**, feeding equipment health monitoring and predictive-maintenance analytics.
- **Built PySpark transformation jobs** on Parquet datasets in HDFS - window functions, aggregations, broadcast joins and schema enforcement - loading partitioned Hive tables.
- **Implemented schema evolution handling** with dynamic schema inference, so new or modified fields from IoT firmware updates flowed through without pipeline redeployment or downstream breakage.
- **Built data validation, lineage tracking and metadata capture** into the ingestion and transformation stages for end-to-end audit traceability.
- **Standardised Spark transformation patterns** into reusable ETL components with product and analytics teams, reducing ad-hoc query load.

***Stack:*** *PySpark, Hadoop, HDFS, Hive, Spark SQL, Parquet, Python, Git, Linux*

### Data Engineer
**Nov 2024 - Aug 2025**

**Tiger Analytics - Bengaluru, India**

*Domain: Automotive*

**Automotive Data Platform - Lakehouse Architecture & ETL Pipeline Optimisation**

- **Implemented Medallion Architecture** (Bronze-Silver-Gold) on Azure Databricks with Delta Lake - raw ingestion through to analytics-ready datasets, with SCD Type 1 & 2 dimension tables using surrogate keys.
- **Cut Spark job runtimes by 35%** through partition pruning, caching, broadcast joins and Adaptive Query Execution (AQE), reducing cluster resource consumption.
- **Built incremental ingestion** with watermark-based change detection using ADF triggers and Airflow DAGs on GCP Cloud Composer, replacing full reloads and reducing data latency.
- **Developed ETL/ELT pipelines** with Apache Spark, Hive and Azure Data Factory for large-scale automotive manufacturing and supply-chain datasets.
- **Automated testing and deployment of Spark jobs** across Dev, QA and Production with Git branching and Jenkins pipelines.

***Stack:*** *Apache Spark, PySpark, Hive, ADF, Azure Databricks, Delta Lake, GCP Cloud Composer, Airflow, Python, Jenkins, Git*

### Associate Data Engineer
**Oct 2020 - Aug 2024**

**XL Dynamics - Mumbai, India**

*Domain: Finance (Mortgage & Loan Processing)*

**Financial Data Processing, ETL Automation & Reporting Platform**

- **Developed ETL pipelines** in Python, SQL, PySpark and Azure Data Factory ingesting and transforming mortgage, loan and risk data from multiple transactional systems - **20M+ financial records** with lineage and audit controls.
- **Implemented data validation rules and reconciliation checks** across pipeline stages, ensuring financial data accuracy from source through to the reporting layer.
- **Wrote Hive and PySpark transformation jobs** for cleansing, deduplication, aggregation and SCD Type 1 & 2 handling on loan and risk dimension tables.
- **Automated file ingestion, scheduling and alerting** with Shell scripts, reducing manual operational effort by **40%**.
- **Produced HLD and LLD design documents** for the finance data platform and built Tableau dashboards for loan-performance and risk KPIs.

***Stack:*** *Python, PySpark, SQL, ADF, Hive, Hadoop, HDFS, Shell Scripting, PostgreSQL, Tableau, Git, Linux*

## CERTIFICATIONS

- Databricks Fundamentals - Databricks
- Build a Data Warehouse with BigQuery - Google Cloud Skill Badge
- Build Batch Data Pipelines on Google Cloud: Cloud Composer - Google Cloud Skill Badge

## EDUCATION

Bachelor of Engineering (B.E.), National Institute of Engineering (NIE), Mysore, 2017 - 2020